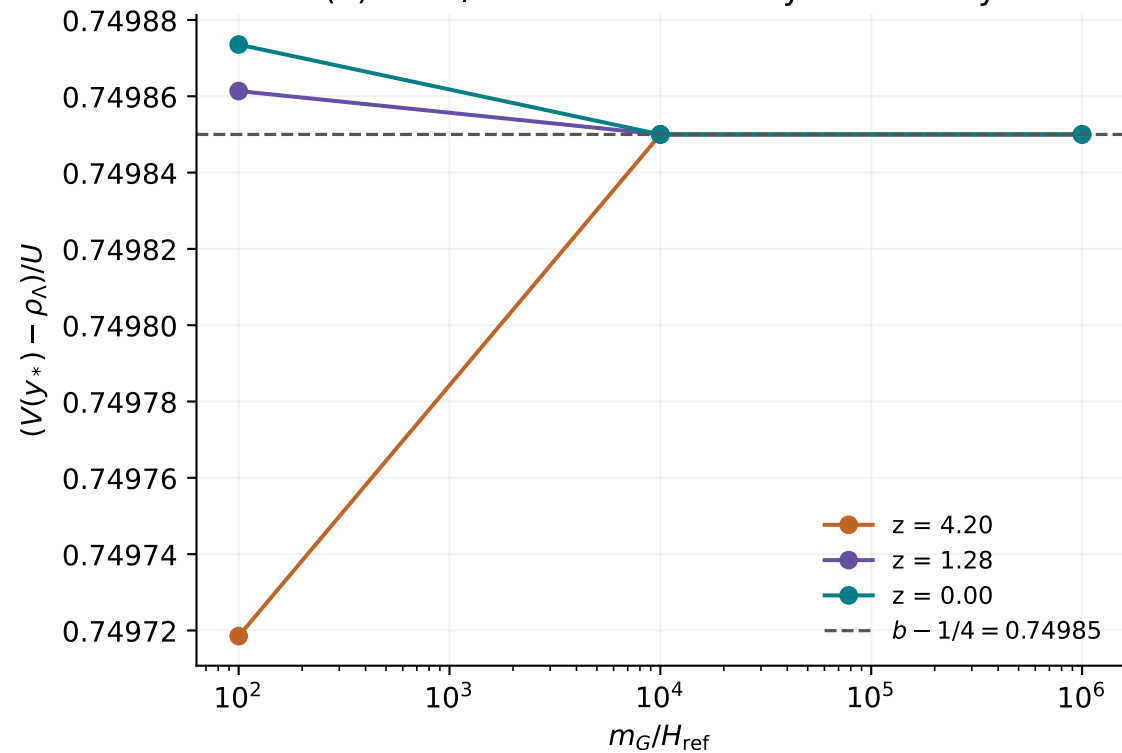


Tree-level exception retained | gauge curve is a partial leading-log diagnostic

(a) $\theta = \pi/2$: a classical heavy-field valley



(b) Direct X-gauge coupling adds a transmission path

